Project Report Day 2

Title: Rule-Based Logic & Intent Classification

TASK-02: Developing a Rule-Based NLP Engine for Live News Streams

Tools Used

- VS Code
- Git & GitHub
- MS Word

Libraries Used

- **requests**: Used to send request to NewsAPI and fetch live news data.
- **dotenv**: Used to load API key securely from .env file.
- **os**: Used to read API key using os.getenv().
- **time**: Used to handle rate limiting by adding delay.
- **re (Regex)**: Used to extract entities like company names, currency, and percentages.
- **json**: Used to store final processed news data into .json file.
- **requests.exceptions**: Used to handle request errors safely.

Errors

1. Incorrect Entity Extraction

Regex was extracting random capital words as companies

Solution:
Used predefined company lists from config.json instead of relying only on regex

2. Request Exception Error

API request failed due to network issues

Solution:
Handled using “from requests.exceptions import RequestException”

3. Missing Data (None values)

Some articles had missing title or description

Solution:
Used : title = article["title"] or ""

Conclusion

This task helped me understand:

- How to fetch real-time news data using APIs
- How to build a rule-based classification system
- How to extract useful information using regex
- How to manage configurations using JSON files
- How to handle API errors and edge cases